

Uncertainty

• Different types of uncertainty effect an agent:

→ state of the world.

→ Effect of actions

Uncertain knowledge of the world:

• Inputs missing

• Limited precision in the sensors

• Incorrect model:

action $\Rightarrow$ state due to the complexity

• A changing world

• Rational decisions

- Relative importance of the sub-goals

- Utility theory

- Degree or belief that sub-goals will be achieved

- Probability theory

Decision theory = Probability theory + utility theory

Handling uncertain knowledge -

- moving beyond rigid
- deterministic logic to probabilistic and
- heuristic framework

Probability

Chances of happening something

1st type $\Rightarrow$ Prior probability

$\downarrow$
independent
unconditional

Before any evidence is found.

$\Rightarrow$ Posterior probability

$\downarrow$
dependent
conditional

After evidence is obtained

* Notation for unconditional probability
for a proposition A : $P(A)$

Sample space

$\downarrow$
All possible events

$\{H, T\}$

$\downarrow \quad \downarrow$
 $E \quad E \quad E = \text{Event}$
 $\quad \quad \quad \cdot 2$

Roll a die :

$\{1, 2, 3, 4, 5, 6\}$

Event = 6

$$P(6) = \frac{1}{6}$$

$$P(A) = \frac{1}{6}$$

$\downarrow \quad \Rightarrow$ Per even 20.71%
Axiom Possibility 1

Probability of
all events

$P(E) \Rightarrow \text{True}$

$P(E^c) = \text{False}$

$\downarrow$
 E^c complement

Axioms for probabilities :

1. $0 \leq P(A) \leq 1$
 2. $P(\text{True}) = 1, P(\text{False}) = 0$
 3. $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$
- Complement $\Rightarrow$ Opposite of an event occurring
 $\Rightarrow$ includes all the possible outcomes that are not the event itself
 $\Rightarrow A^c, A'$ or not A

$$P(E) + P(E^c) = 1$$

$$\Rightarrow P(E) = 1 - P(E^c)$$

$$\Rightarrow P(E^c) = 1 - P(E)$$

- Mutually Exclusive Events

- $\rightarrow$ disjoint आखण्ड

- $\rightarrow$ Match आखण्ड ना

- $\rightarrow$ Independent इकल .

Only one of them
are true

- Random variable

- Has a domain of possible values

- Each value has a assigned probability between 0 and 1

Probability Model

$$\Omega \rightarrow \{1, 2, 3, 4, 5, 6\}$$

$$\omega \rightarrow \{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}$$

Conditional Probability

→ একটি event এর পর অন্য একটি event হওয়ার probability.

Notation: $P(E|F)$ → Conditional probability that E occurs given that, F has occurred

$$P(E|F) = \frac{P(E \cap F)}{P(F)} \quad \text{if } P(F) \neq 0$$

$$P(E) = P(F|E) P(E)$$

Ex Cards numbered one through ten are placed in a hat. one card is drawn. Given that if number on the drawn card is at least five, then what is the probability that, it is ten?

⇒ E: the card is ten

F: the card is at least 5. $\rightarrow P(E \cap F) = 1$

at least 5
so, 5 to 10
sample space:
1-10

$$\therefore P(E|F) = \frac{\frac{1}{10}}{\frac{6}{10}} = \frac{1}{6}$$